MATERIALS & CONDENSED MATTER CONCENTRATION

(4) Applied Physics Core Courses + (6) MCM Electives, including at least (3) 700-level courses

CORE COURSES

- 615.641 Mathematical Methods for Physics & Engineering — Every Term • Asynchronous
- 615.642 Electromagnetics — FA / SP • Asynchronous
- 615.651 Statistical Mechanics and Thermodynamics — Every Term • Asynchronous
- 615.680 Materials Science — FA / SP • Asynchronous

Unshaded boxes indicate elective courses not offered by the Applied Physics program. Check the EP website / SIS for course offering information.

Electives

- 615.654 Quantum Mechanics — FA / SP • Asynchronous
- 615.682 Materials Characterization — In Development • Async.
- 615.683 Materials in Extreme Environments — In Development • Async.
- 515.617 Nanomaterials
- 515.635 Mechanical Properties of Materials
- 525.606 Electronic Materials
- 525.621 Introduction to Electronics & the Solid State
- 535.684 Modern Polymeric Materials

700-level courses – Select at least 3

- 615.715 Applied Atomic and Molecular Physics — Spring • Asynchronous
- 615.747 Sensors and Sensor Systems — FA / SP • Asynchronous
- 615.757 Solid State Physics — FA • Asynchronous
- 615.760 Physics of Semiconductor Devices — SP • Asynchronous
- 615.763 Infrared Sensors: Physics and Applications — SP • Asynchronous
- 615.784 Superconducting Devices: Physics and Applications — FA / SU • Asynchronous
- 615.800 Applied Physics Project — Every Term • Asynchronous
- 615.802 Directed Studies in Applied Physics — Every Term • Asynchronous
- 535.732 Fatigue and Fracture of Materials
- 535.748 Stress Waves, Impacts, and Shockwaves